

SIDHANT CHANDRAKAR

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Professional Summary

Detail-oriented M.Tech Computer Science student specializing in systems design, data analytics, and networking. Proven hands-on experience building machine learning architectures, programmable network security prototypes, and containerized applications. Seeking to leverage strong analytical frameworks, dynamic backend development, and Python automation in technical and analytical roles.

Education

Visvesvaraya National Institute of Technology (VNIT) , Nagpur, Maharashtra	2025 – 2027
<i>Master of Technology (M.Tech) in Computer Science and Engineering</i>	CGPA: 8.73
Chhattisgarh Swami Vivekanand Technical University , Bhilai, Chhattisgarh	2020 – 2024
<i>Bachelor of Technology (B.Tech) in Information Technology</i>	CGPA: 8.38

Technical Skills

- **Programming Languages:** Python, C, C++
- **Tools & Frameworks:** PyTorch, Docker, Git, Linux Terminal, Flask, VS Code
- **Networking & Security:** P4, Mininet, BMv2, Wireshark
- **Databases:** Oracle, Amazon Redshift, Cassandra

Research Experience

Hyperspectral Image Classification Architecture [GitHub Link](#)

Tools Used: PyTorch, Python, Parallel Feature Extraction Frameworks

- **Problem Solved & Workflow:** Remote sensing data processing suffers from extreme data redundancy and computational bottlenecks. Designed a novel HSI classification architecture using parallel spectral-spatial feature extraction and Bidirectional MamTrans implemented via PyTorch.
- Integrated dynamic token compression mechanisms to automatically prune redundant data tokens during execution, drastically reducing model training times and inference execution overhead while maintaining high accuracy across benchmark datasets.

Projects

Stock Market Analytics Dashboard [GitHub Link](#)

Tools Used: Python, Upstox API, PyTorch, Pandas, Websockets

- **Problem Solved & Workflow:** Financial market tracking often suffers from latency and complex raw datasets, making real-time analysis difficult.
- Ingested live stock data directly via the Upstox API into a custom Python data pipeline for cleaning and automated feature engineering.

P4-Based DDoS Detection System [GitHub Link](#)

Tools Used: P4, Mininet, BMv2, Python, Wireshark

- **Problem Solved & Workflow:** Traditional software-based DDoS defenses introduce significant CPU overhead and high processing delays because attack packets must reach central host servers.
- Built a programmable data-plane security prototype using P4 deployed on BMv2 switches in a simulated Mininet network.
- Configured the in-network workflow to parse packet headers and track flow counts directly at the switch level to detect abnormal traffic bursts in real-time.

Hybrid Search & Re-Ranking Engine [GitHub Link](#)

Tools Used: Python, PyTorch, Sentence-Transformers, BM25, FAISS

- **Problem Solved & Workflow:** Single-mode retrieval systems suffer from vocabulary mismatch in keyword search or lack of contextual precision in semantic vector search.
- Implemented a hybrid RAG retrieval pipeline combining BM25 sparse keyword search and dense vector embeddings to maximize overall search recall.
- Integrated a cross-encoder re-ranking stage to re-score candidate items, significantly improving contextual precision for accurate downstream generation.